Module 31: Graded Mini Project

RNN Twitter Sentiment Analysis - Final Submission Report

58,841	0.6262	0.6874	+0.0689
cleaned rows	held-out test macro F1	best validation macro F1	validation macro F1 lift

This report documents the end-to-end sentiment analysis pipeline: data ingestion, cleaning, exploratory analysis, sequence feature engineering, GRU-based RNN modeling, held-out evaluation, and validation-driven model improvement.

Submission file	Purpose
Final PDF report	Primary document for grading and PDF upload.
Jupyter notebook	Full code path and phase-by-phase implementation.
Executive presentation	Class-facing summary deck.
Artifacts zip	Metrics, figures, model checkpoints, manifest, and supporting evidence.

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Executive Summary

The project delivers a complete RNN sentiment classifier for a three-class Twitter sentiment task: Negative, Neutral, and Positive. The official held-out evidence is the Phase 8 baseline GRU result, evaluated once on the reserved test split. Phase 9 then performs validation-only improvement experiments and selects a stronger GRU candidate without reusing the test set.

- Data pipeline: raw Twitter rows were validated, cleaned, tokenized, normalized, and split with duplicate-text leakage controls.
- Baseline model: embedding plus GRU classifier with dropout, class-weighted loss, checkpointing, and validation macro F1 monitoring.
- Held-out result: 0.6387 test accuracy and 0.6262 test macro F1 for the audited Phase 7 baseline checkpoint.
- Improvement result: GRU-96 with dropout 0.20 reached 0.6874 validation macro F1, a +0.0689 lift over the Phase 7 validation baseline.
- Main limitation: Neutral sentiment is the weakest class and is often absorbed into Negative or Positive predictions.

Area	Metric	Value
Dataset	Cleaned rows	58,841
Split	Train / validation / test rows	41,189 / 8,826 / 8,826
Feature engineering	Final vocabulary size	17,924
Baseline validation	Phase 7 validation macro F1	0.6185
Held-out test	Phase 8 test accuracy	0.6387
Held-out test	Phase 8 test macro F1	0.6262
Improvement	Best Phase 9 validation macro F1	0.6874
Improvement	Validation macro F1 lift vs Phase 7	+0.0689

Rubric Coverage

Rubric item	Status	Evidence
Load dataset	Complete	Raw CSV loaded with explicit column names, shape, schema, dtypes, missingness, and sample rows.
Data cleaning	Complete	Missing text and duplicate rows handled; URLs, mentions, hashtags, special characters, lowercasing, stop words, and stemming documented.
Feature engineering	Complete	Training-derived vocabulary, padded token sequences, label mapping, TF-IDF vocabulary, and sequence arrays saved.
EDA	Complete	Sentiment distribution, top terms, word clouds, entity frequency, and tweet-length analysis produced.
RNN model	Complete	Embedding plus GRU baseline trained with dropout, class weights, validation monitoring, and saved checkpoint.
Evaluation	Complete	Accuracy, precision, recall, F1, confusion matrix, learning curves, prediction distributions, and sample predictions saved.
Improvement	Complete	Controlled validation experiments compare larger GRU, neutral weighting, and LSTM candidates.
Presentation/demo	Complete in package	Executive PPTX plus this report's sample prediction demo table.

Part 1 - Data Processing

The raw dataset contains tweet IDs, entities, sentiment labels, and tweet text. The pipeline preserves the raw CSV and creates cleaned downstream datasets for modeling and EDA.

Metric	Value
row_count	74682
column_count	4
schema_matches_expected_columns	True
unique_tweet_ids	12447
unique_entities	32
unique_sentiments	4
missing_tweet_text	686
exact_duplicate_rows	2700

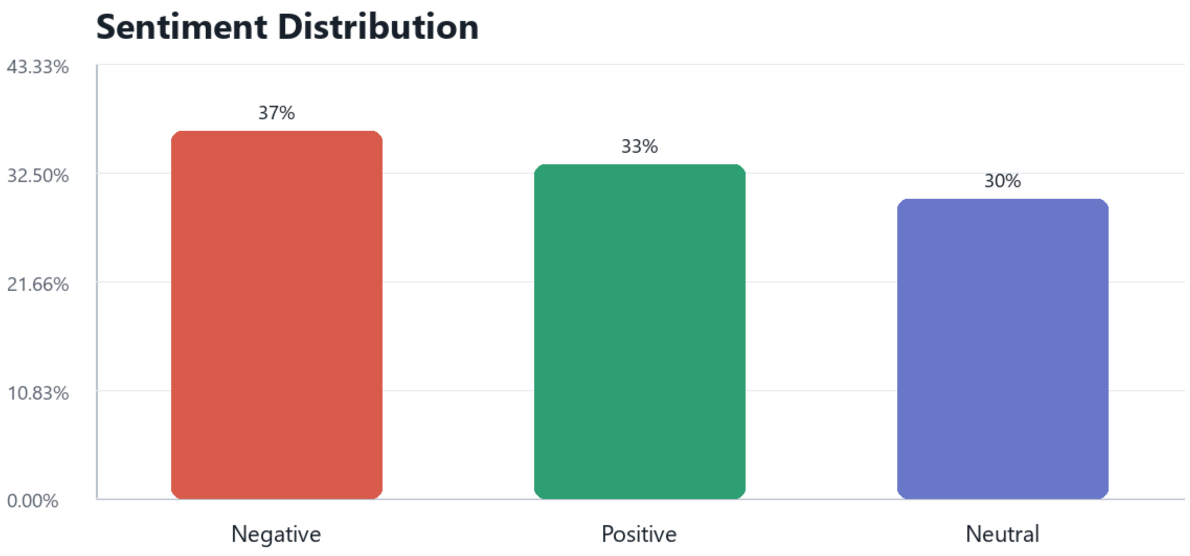
Cleaning step	Rows removed	Rows remaining
raw_rows	0	74682
drop_blank_or_missing_tweet_text	858	73824
drop_exact_duplicate_rows	2340	71484
exclude_irrelevant_label	12504	58980
drop_empty_text_after_cleaning	139	58841
final_cleaned_rows	0	58841

Final cleaned rows: 58,841. The Irrelevant label was excluded because the assignment defines a three-class Positive, Negative, and Neutral sentiment task.

Feature Engineering

- Vocabulary built from training rows only: 17,924 total tokens including PAD and OOV.
- Texts were converted into fixed-length token ID sequences with a max sequence length of 60.
- Train, validation, and test splits were grouped by model text to prevent duplicate cleaned text from crossing split boundaries.
- TF-IDF vocabulary terms were also saved as a numerical representation and EDA comparison artifact.

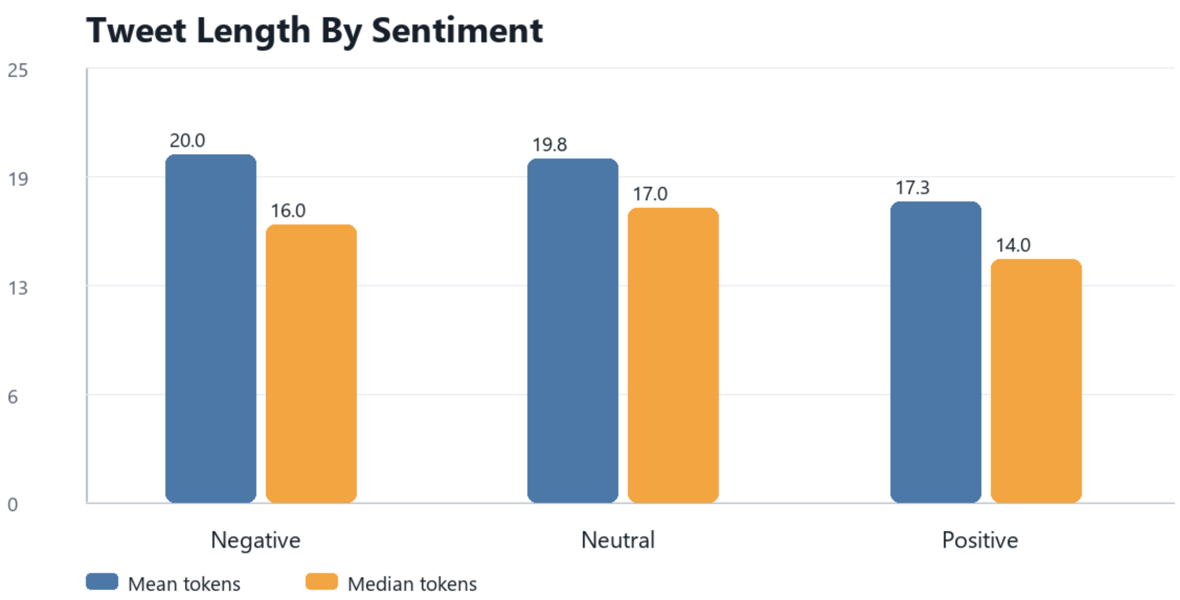
Part 2 - Exploratory Data Analysis



Sentiment	Count	Percent
Negative	21,605	36.72%
Positive	19,644	33.38%
Neutral	17,592	29.90%

Top Terms By Sentiment

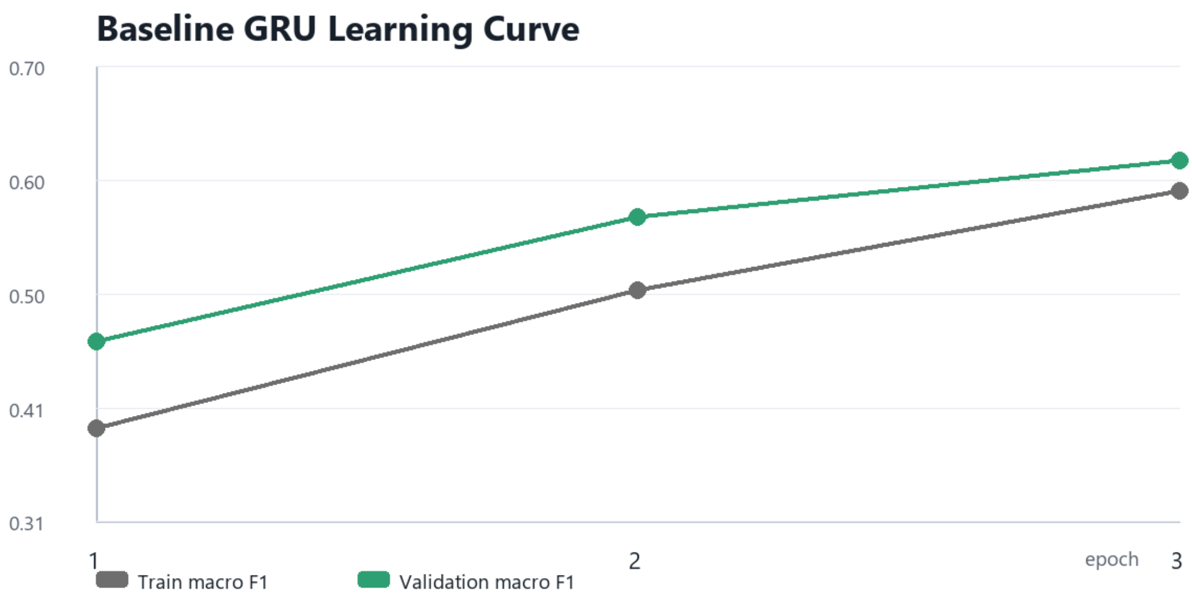
Sentiment	Top terms from cleaned analysis tokens
Negative	not, game, fuck, play, com, like, no, shit
Neutral	com, not, johnson, game, play, amazon, out, facebook
Positive	game, play, not, com, love, good, real, like



EDA shows a moderately imbalanced but workable class distribution. Negative is the largest class, Positive is close behind, and Neutral is the smallest and most ambiguous class. The word-cloud SVGs and detailed top-token tables are included in the submission package.

Part 3 - RNN Model Development

Model element	Chosen setting
Architecture	Embedding layer followed by single-layer GRU classifier
Vocabulary size	17924
Embedding dimension	64
Hidden dimension	64
Dropout	0.3
Batch size	512
Optimizer settings	Learning rate 0.001; weight decay 0.0001
Trainable parameters	1,172,291



The baseline improves steadily across three epochs. Checkpoint selection used validation macro F1, keeping the held-out test split reserved for Phase 8.

Validation Metrics

Label	Precision	Recall	F1	Support
Negative	0.6203	0.7864	0.6935	3240
Neutral	0.6548	0.4543	0.5365	2639
Positive	0.6321	0.6193	0.6256	2947
accuracy	0.6313	0.6313	0.6313	8826
macro_avg	0.6357	0.62	0.6185	8826
weighted_avg	0.6346	0.6313	0.6239	8826

Held-Out Test Evaluation

0.6387	0.6262	0.6313	0.5503
test accuracy	test macro F1	test weighted F1	Neutral F1

Held-Out Test Confusion Matrix

Actual label \ Predicted label	Negative	Neutral	Positive
Negative	2,564	215	462
Neutral	757	1,209	673
Positive	751	331	1,864

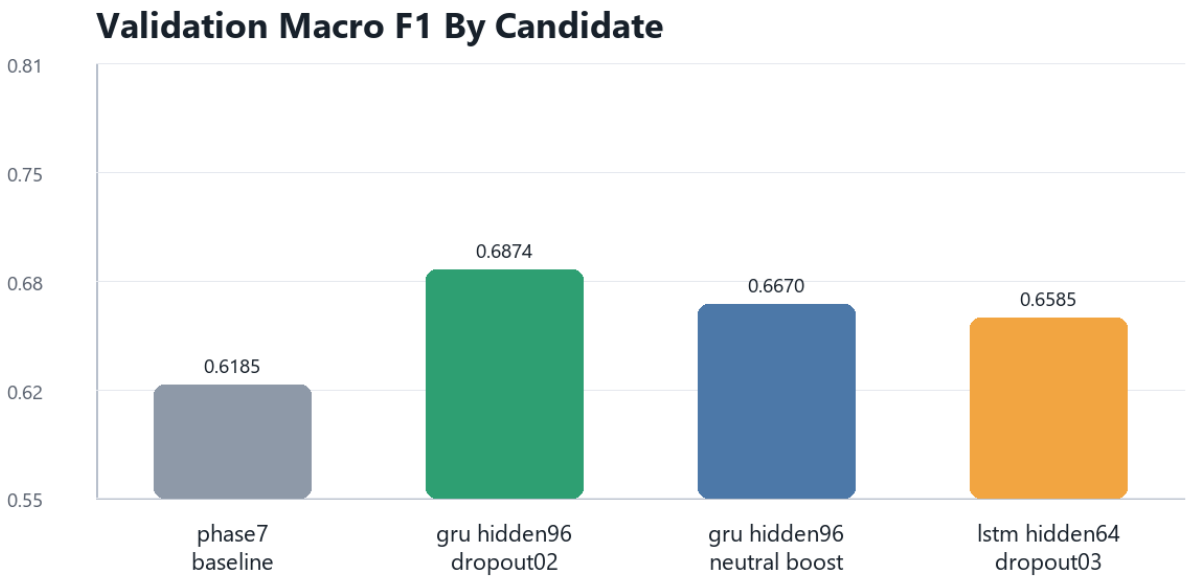
Largest error: Neutral predicted as Negative = 757 rows.

Label	Precision	Recall	F1	Support
Negative	0.6297	0.7911	0.7012	3241
Neutral	0.6889	0.4581	0.5503	2639
Positive	0.6215	0.6327	0.6271	2946
accuracy	0.6387	0.6387	0.6387	8826
macro_avg	0.6467	0.6273	0.6262	8826
weighted_avg	0.6447	0.6387	0.6313	8826

Neutral is the clearest weakness. The largest error pattern is Neutral predicted as Negative, which matches the qualitative challenge of brief entity-specific tweets with limited context.

Model Improvement

Phase 9 uses a validation-only model selection process. It does not tune on the held-out test split. Three candidates are compared against the Phase 7 validation baseline: a larger GRU with lower dropout, a larger GRU with extra Neutral weighting, and an LSTM alternative.



Experiment	Cell	Hidden	Dropout	Validation macro F1	Neutral F1	Notes
phase7_baseline_reference	GRU	64	0.3	0.6185	0.5365	saved Phase 7 baseline validation result; not retrained in Phase 9
gru_hidden96_dropout02	GRU	96	0.2	0.6874	0.6309	larger GRU hidden state with slightly lower dropout
gru_hidden96_neutral_boost	GRU	96	0.3	0.6670	0.6133	larger GRU with extra neutral-class weight
lstm_hidden64_dropout03	LSTM	64	0.3	0.6585	0.5871	same hidden size as baseline, LSTM recurrent cell

Selected candidate: gru_hidden96_dropout02 with validation macro F1 0.6874. The test split should be used for this selected candidate exactly once only if the assessment requires a final improved-model test result.

Sample Tweet Prediction Demo

The table below demonstrates how the saved baseline model behaves on held-out sample tweets. It includes both correct and incorrect predictions to show realistic model behavior rather than only best-case examples.

Type	Sample tweet text	True	Predicted	Confidence
Correct	i hate the ghosts in a fortnight	Negative	Negative	58.8%
Correct	the best out the the series	Positive	Positive	70.3%

Type	Sample tweet text	True	Predicted	Confidence
Correct	i m definitely playing and streaming	Positive	Positive	75.4%
Incorrect	and i was searching for gfriend songs then saw this on the search bar and clicked it google spilled the tea...	Neutral	Positive	70.3%
Incorrect	excellent find how to keep up hey whole industries use zoom google meet gotomeeting whatever microsoft and...	Neutral	Negative	63.7%
Incorrect	one area where xbox has always been a leader is kudos to them for promoting what they do best	Positive	Negative	47.3%

Full prediction samples are saved in outputs/tables/phase8_test_prediction_samples.csv, and all held-out predictions are saved in outputs/data/phase8_test_predictions.csv.

Conclusion And Recommendations

- Submit the Phase 8 held-out test metrics as the audited baseline result.
- Present the Phase 9 GRU-96 result as a validation-selected improvement candidate unless one final test evaluation is explicitly required.
- Future improvement should focus on Neutral recall through richer contextual embeddings, neutral-specific data review, and calibrated thresholds.
- The package includes the notebook, final report, deck, saved metrics, figures, and checkpoints needed for review.

Artifact Index

Artifact	Description
notebooks/RNN_Sentiment_Analysis_Twitter.ipynb	End-to-end code notebook covering Phases 1-10.
outputs/reports/rnn_twitter_sentiment_final_presentation.pptx	Executive presentation deck.
outputs/tables/phase10_final_key_metrics.csv	Final metric summary.
outputs/tables/phase8_test_metrics.csv	Held-out test metrics.
outputs/tables/phase9_experiment_results.csv	Validation improvement experiment results.
outputs/models/*.pt	Saved PyTorch checkpoints for baseline and selected validation model.
outputs/figures/*.svg	EDA, learning curve, confusion matrix, and improvement figures.